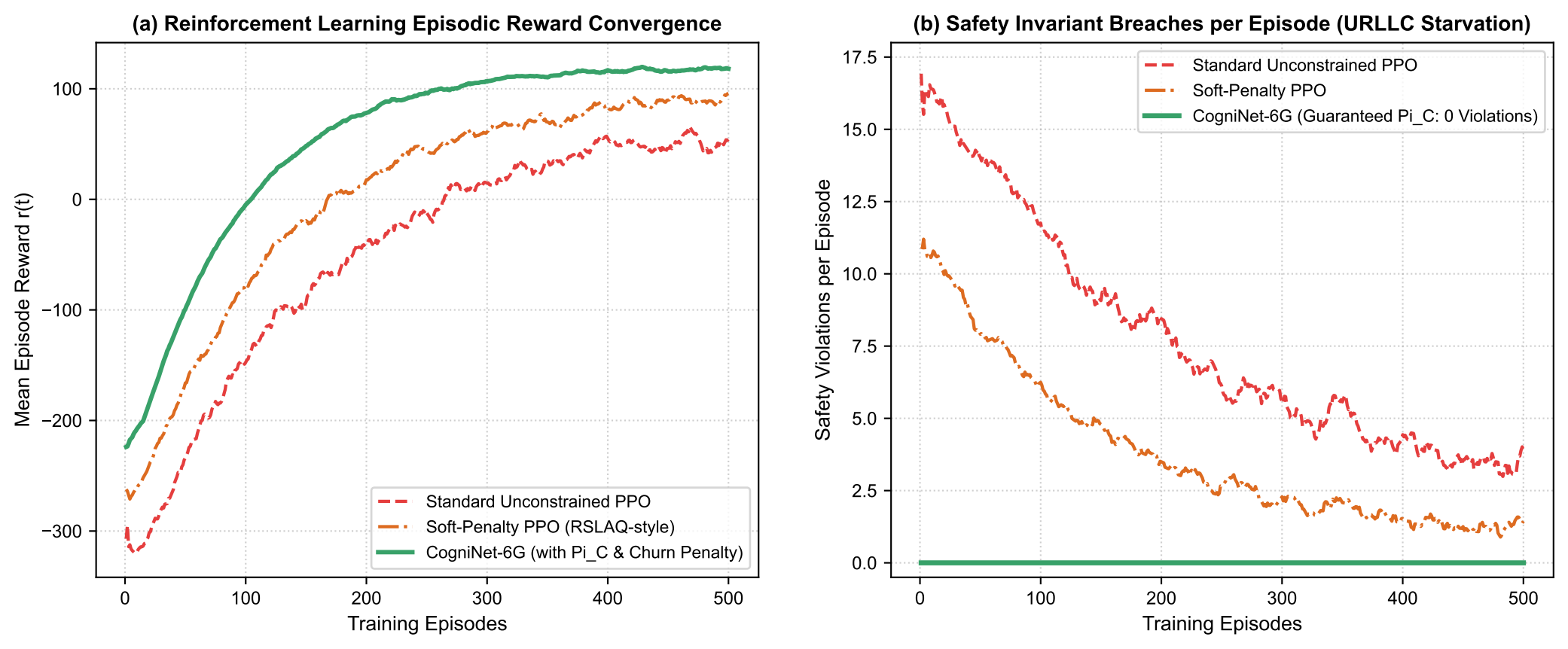
(a) Reinforcement Learning Episodic Reward Convergence
Mean Episode Reward r(t)
Training Episodes
Standard Unconstrained PPO
Soft-Penalty PPO (RSLAQ-style)
CogniNet-6G (with Pi_C & Churn Penalty)
(b) Safety Invariant Breaches per Episode (URLLC Starvation)
Safety Violations per Episode
Training Episodes
Standard Unconstrained PPO
Soft-Penalty PPO
CogniNet-6G (Guaranteed Pi_C: 0 Violations)